DATA CONNECTION GUIDE / VERSION 2.1.0.0

Populate your Finance OS dashboard

One Power BI visual. Five internal pages.
A blank workspace ready for your own ledger.

This release contains no bundled records.
The visual has no sample-data fallback. The companion model is a typed empty table, and the CSV template contains headers only. Metrics appear only after you supply and bind valid ledger rows.

Choose your starting point

File	Use it for
FinanceOS.pbip	Open the preconfigured blank project. It has one report page and one Finance OS visual, with all fields already mapped.
Finance-OS.pbiviz	Import the empty visual into an existing Power BI report and map the ledger fields yourself.
FinanceLedger-Template.csv	Populate this header-only file with your event ledger. Keep dates in yyyy-MM-dd format.
Connect-FinanceLedger.pq	Replace the blank FinanceLedger query with a CSV connection. Change the placeholder path first.

The shortest route

Fill the CSV, open FinanceOS.pbip, replace the FinanceLedger query with the supplied Power Query script, change its file path, then Close & Apply. The existing visual bindings are retained when the table and column names stay unchanged.

Before you begin

Have a recent Power BI Desktop version, permission to use a private visual, a single reporting currency, and access to complete invoice and payment history. You do not need Node.js or a separate web server to use the packaged visual.

Scope: this document covers the current single-visual Finance OS application. Earlier report/demo versions are not the data source for this blank template.

01 / PREPARE THE SOURCE

Build the event ledger

Each row represents one dated business event. The visual rebuilds month-end balances from those events; it does not consume a pre-aggregated KPI table or repeated monthly balance snapshots.

EventType	EventDate	Amount means	Required rule
Invoice	Invoice issue date	Original invoice value	Exactly one Invoice event for each InvoiceID.
Payment	Cash allocation date	Amount allocated to this invoice	Use a separate row for each partial payment or allocation.
Credit	Credit effective date	Credit allocated to this invoice	Supply a positive credit value; the visual subtracts it from AR.
Promise	Promise creation date	Amount promised by PromiseDue	PromiseDue must be populated. The promise itself does not reduce AR.

Join the invoice attributes onto every event

Use InvoiceID to attach Customer, original InvoiceDate, DueDate and the optional dimensions to payment, credit and promise rows. Keep these values consistent across an invoice lifecycle. The original Invoice event supplies the attributes used for grouping.

Rules that prevent duplicated balances

1. Make EventID globally unique. Prefix source keys by event type if different systems reuse IDs.
2. Do not repeat the original invoice amount on Payment or Credit rows. Amount is the amount of that event.
3. Allocate a multi-invoice bank receipt by invoice; do not repeat the full bank receipt on every allocation.
4. Do not append monthly AR snapshots as new Invoice events.
5. Retain the original Invoice row and all subsequent events needed to reconstruct balances.
6. Use a unique Customer label, such as account code plus name, if account names are not unique.
7. Convert currencies upstream. The visual only formats one reporting currency; it does not calculate FX.
8. Keep IDs as text, even if they contain only digits, so leading zeros are preserved.

Keep complete lifecycles together.
Customer, invoice and security filters must retain the relevant Invoice and event rows. A report-level EventDate filter can remove an old invoice while retaining its recent payments. Use the visual's Reporting close selector for historical analysis instead.

02 / DATA DICTIONARY

Map the fields exactly

Drag each model column into the matching visual data well. The eight fields marked Required are mandatory. Optional columns can be left blank; the supplied query also adds absent optional columns.

Visual field	CSV / model column	Power BI type	Requirement / interpretation
Event ID	EventID	Text	Required. Unique event key across the full ledger.
Invoice ID	InvoiceID	Text	Required. Stable invoice key repeated on its events.
Customer	Customer	Text	Required. Unique customer/account label.
Invoice date	InvoiceDate	Date	Required. Original invoice issue date.
Due date	DueDate	Date	Required. Date that invoice payment becomes due.
Event date	EventDate	Date	Required. Date of this event, not the reporting month.
Event type	EventType	Text	Required. Invoice, Payment, Credit or Promise, with this capitalization.
Event amount	Amount	Fixed decimal	Required. Select Sum of Amount in the data well.
Region	Region	Text	Optional regional grouping.
Segment	Segment	Text	Optional customer segment.
Collector	Collector	Text	Optional invoice owner/collector.
Risk tier	RiskTier	Text	Optional risk grouping; for example your governed risk categories.
Dispute date	DisputeDate	Date	Optional. On the Invoice event, date the entire open invoice became disputed.
Promise due	PromiseDue	Date	Required on Promise events; otherwise blank.

Formatting is not data transformation.
Use ISO dates such as yyyy-MM-dd in the CSV. Put raw numeric values in Amount, without currency symbols. Keep EventID in the visual so Sum of Amount is evaluated per unique event. The visual's currency setting changes the display label, not the underlying currency.

03 / CONFIGURE THE REPORT

Load and connect in Power BI

Recommended: use the included blank project

01 Extract the whole package
Keep FinanceOS.pbip beside the FinanceOS.Report and FinanceOS.SemanticModel folders. Open FinanceOS.pbip in Desktop. If your version needs project/PBIR support enabled, enable it and reopen the project.

02 Replace the existing blank query
Choose Transform data. In Power Query select the existing FinanceLedger query, then Advanced Editor. Replace its content with Connect-FinanceLedger.pq. Do not create a second table named FinanceLedger2.

03 Point it at your populated file
Change LedgerCsvPath in the first lines of the supplied script to your populated CSV. The script checks the eight required column names, adds missing optional columns, trims text and applies date/amount types.

```
LedgerCsvPath = "C:\Data\FinanceLedger.csv",
```

04 Apply and inspect
Confirm the query preview contains your rows and no conversion errors, then choose Close & Apply. Keep the table name FinanceLedger and the documented column names so the prebound visual continues to resolve its fields.

05 Confirm the visual bindings
Select the Finance OS visual. Check the 14 field mappings against page 3. Keep EventID and use Sum of Amount. Set the reporting currency in Format visual > Finance workspace.

Alternative: add it to an existing report

Load your ledger with Get data > Text/CSV (or your approved ERP/database connector), then Transform Data and set its types. In the Visualizations pane choose the ellipsis > Import a visual from a file, select Finance-OS.pbiviz, add it to the page and map the fields from page 3. A canvas of roughly 1,200 x 850 or larger works well; smaller sizes use the responsive sidebar and scrolling.

Local visual imports may require developer mode.
If required by your Desktop version, open File > Options and settings > Options > Current file > Report settings, enable Develop a visual, and select OK. This is a session setting. Your organization must allow private custom visuals. Microsoft reference links are on page 6.

04 / RECONCILE BEFORE RELYING ON TOTALS

Validate the first reporting close

A populated chart is not proof that the ledger is complete. Reconcile a small, known invoice population first, then expand to the full portfolio.

Control	What to verify
Identity	One Invoice event per InvoiceID; no duplicate EventID values; every Payment/Credit/Promise references an invoice that is present.
History	Dates are valid and complete through the reporting close. No report filter or security rule leaves orphan events.
Balance	For each invoice at the close: original amount minus effective payments minus effective credits equals outstanding balance.
Portfolio	Sum positive outstanding invoice balances to AR. Reconcile unapplied cash, credit balances and other excluded items separately.
Flow	Monthly cash is the month's Payment allocations, not cumulative payments. Promises never count as cash.
Aging	An invoice due on the close is Current. Past-due buckets are 1-30, 31-60, 61-90 and over 90 days.

Worked check - illustrative only, not supplied data

An invoice of 1,000 is issued on 1 January and due on 31 January. A payment of 400 is allocated on 15 February; a credit of 100 is effective on 5 March. The expected closing balances are:

Reporting close	Calculation	Expected open AR
31 January	1,000 - 0 - 0	1,000
28 February	1,000 - 400 - 0	600
31 March	1,000 - 400 - 100	500

Check the ratios separately

DSO = closing AR / credit invoice sales in the preceding 90 calendar days x 90. CEI = (opening AR + monthly billings - credits - ending AR) / (opening AR + monthly billings - credits - ending current AR). A zero denominator displays an em dash. The first two loaded months may lack a complete 90-day sales window.

Treat data-reduction warnings as incomplete totals.
This visual requests up to 30,000 event rows and does not fetch additional segments. Narrow the source to a complete customer/invoice population or redesign the source before using portfolio totals. Do not trim individual event types or dates to fit the limit.

05 / MAINTAIN A RELIABLE WORKSPACE

Operate, refresh and troubleshoot

Daily use

The sidebar switches between five pages inside the same visual. Reporting close, region, segment, collector, risk and search filters are shared during the session. Select a month on a chart, a region or an aging bucket to drill into the relevant population. Tables support sorting and pagination; invoice links open an activity drawer. The explicit invoice selection/context-menu actions use the Power BI host, at the source invoice-event row grain.

Refreshing the source

Editing a CSV alone does not refresh the model. Save the source file, then select Refresh in Desktop and confirm the latest reporting month. For a published Import model, configure the source credentials, any required gateway and refresh schedule in the Power BI service. A local C: path is not automatically accessible to the service. The empty PBIP query deliberately reads no file until you replace it.

Symptom	Action
Connect your finance ledger	No fields are bound. Load the ledger and map all eight required fields.
No ledger rows / no periods	The bound table is empty or filtered out. Check Power Query row counts and report/security filters.
Missing required fields	Map the named data wells. A similar display name does not substitute for an actual binding.
Malformed rows / date errors	Use valid dates and numeric Amount values. On Promise events supply PromiseDue. Read the visible warning before using totals.
Wrong AR / duplicate invoices	Remove repeated snapshots, summary rows and duplicated payment totals. Reconcile a single invoice lifecycle.
Unexpected zero cash or stale month	Check Payment event dates, refreshed source rows and the Reporting close selector; cash is monthly.

Implementation limits

One reporting currency; no FX engine, write-off events, unapplied cash or operational write-back. The fixed operating references are a 45-day DSO target, 25% overdue share and cash target equal to 95% of monthly billings. Review these illustrative references before using the dashboard as a performance scorecard. Currency, title, accent, default theme and opening page are formatting controls.

Microsoft references

Import a visual from a file | Developer mode
Text/CSV connector | Data refresh

Guide checked against the visual's source contract and Microsoft documentation on 22 October 2026. This release passes source/build checks; Desktop refresh, host interactions and service refresh must be verified in your environment.